REVIEW - EXAM 2

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Name: Solutions

Approximate Integration

Use the function $f(x) = 3\pi \sin(2x)$ to answer Problems 1 and 2.

1. Use the inequality

$$|E_M| \leq \frac{K(b-a)^3}{24n^2}$$

where $\left|f''(x)\right| \leq K$ on $[a, b]$, to find the number of intervals needed to estimate

$$\int_0^{\frac{\pi}{2}} f(x)\,\mathrm{d}x$$

using the Midpoint Rule with an error less than 10^{-4}. You may use $\pi^2 \approx 10$ to make the computation simpler.

$f'(x) = 6\pi \cos(2x)$

$f''(x) = -12\pi \sin(2x)$

$|f''(x)| = |-12\pi \sin(2x)|$

$= 12\pi |\sin(2x)|$

$\leq 12\pi$

$|E_M| < \frac{12\pi(\pi/2 - 0)^3}{24n^2} = \frac{\pi(\pi^3/8)}{2n^2} = \frac{\pi^4}{16n^2} < \frac{1}{10^4}$

$\Rightarrow \frac{10^4\pi^4}{16} < n^2$

$\Rightarrow \frac{10^2\pi^2}{4} = 25\pi^2 \approx \boxed{250 < n}$

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2. Use the Midpoint Rule with $n = 2$ intervals to estimate the value of the integral

$$\int_0^{\frac{\pi}{2}} f(x)\,\mathrm{d}x.$$

$$\Delta x = \frac{\frac{\pi}{2} - 0}{2} = \frac{\pi}{4}$$

$$x_0 = 0, \quad x_1 = \pi/4, \quad x_2 = \frac{\pi}{2}$$

$$\bar{x}_1 = \frac{\frac{\pi}{4} + 0}{2} = \frac{\pi}{8}$$

$$\bar{x}_2 = \frac{\pi/2 + \frac{\pi}{4}}{2} = \frac{\frac{3\pi}{4}}{2} = \frac{3\pi}{8}$$

$$M_2 = \left[f(\bar{x}_1) + f(\bar{x}_2)\right]\Delta x$$

$$= \left[3\pi \sin\left(2\left(\frac{\pi}{8}\right)\right) + 3\pi \sin\left(2\left(\frac{3\pi}{8}\right)\right)\right]\frac{\pi}{4}$$

$$= \left[3\pi \sin\left(\frac{\pi}{4}\right) + 3\pi \sin\left(\frac{3\pi}{4}\right)\right]\frac{\pi}{4}$$

$$= \left[3\pi \frac{\sqrt{2}}{2} + 3\pi \frac{\sqrt{2}}{2}\right]\frac{\pi}{4}$$

$$= 2\left(\frac{3\pi\sqrt{2}}{2}\right)\frac{\pi}{4}$$

$$= \boxed{\frac{3\sqrt{2}\pi}{4}}$$

Differential Equations

Solve the initial value problems.

3. $y' = xe^y$, $y(0) = 0$

$$\frac{dy}{dx} = xe^y \Rightarrow e^{-y}dy = x\,dx$$
$$\Rightarrow \int e^{-y}dy = \int x\,dx$$
$$\Rightarrow -e^{-y} = \frac{1}{2}x^2 + C$$
$$\Rightarrow e^{-y} = -\frac{1}{2}x^2 + C$$
$$\Rightarrow -y = \ln\left(C - \frac{1}{2}x^2\right)$$
$$\Rightarrow y = -\ln\left(C - \frac{1}{2}x^2\right)$$

$$0 = -\ln\left(C - \frac{1}{2}(0)^2\right) = -\ln(C)$$
$$\Rightarrow C = 1$$
$$\boxed{y = -\ln\left(1 - \frac{1}{2}x^2\right)}$$

4. $P' = \sqrt{Pt}$, $P(1) = 2$.

$$\frac{dP}{dt} = \sqrt{P}\sqrt{t}$$
$$\Rightarrow P^{-1/2}dP = t^{1/2}dt$$
$$\Rightarrow \int P^{-1/2}dP = \int t^{1/2}dt$$
$$\Rightarrow 2\sqrt{P} = \frac{2}{3}t^{3/2} + C$$
$$\Rightarrow \sqrt{P} = \frac{1}{3}t^{3/2} + C$$
$$\Rightarrow P = \left(\frac{1}{3}t^{3/2} + C\right)^2$$

$$2 = \left(\frac{1}{3} + C\right)^2$$
$$\Rightarrow C = -\frac{1}{3} + \sqrt{2}$$
$$\boxed{P = \left(\frac{1}{3}t^{3/2} - \frac{1}{3} + \sqrt{2}\right)^2}$$

Parametric Equations

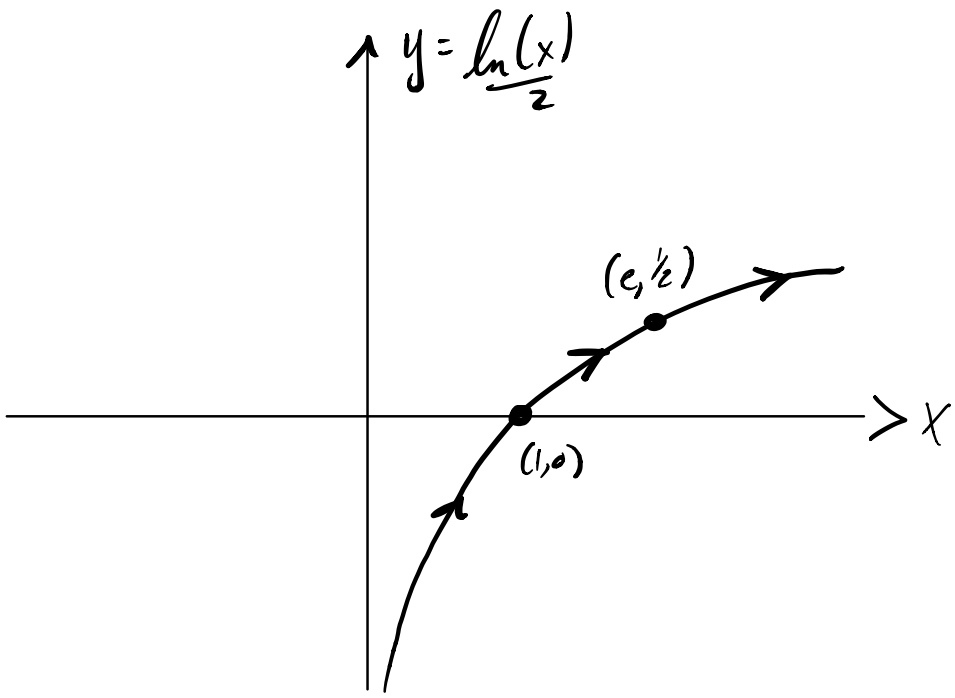
5. Eliminate the parameter to find a Cartesian equation for the curve

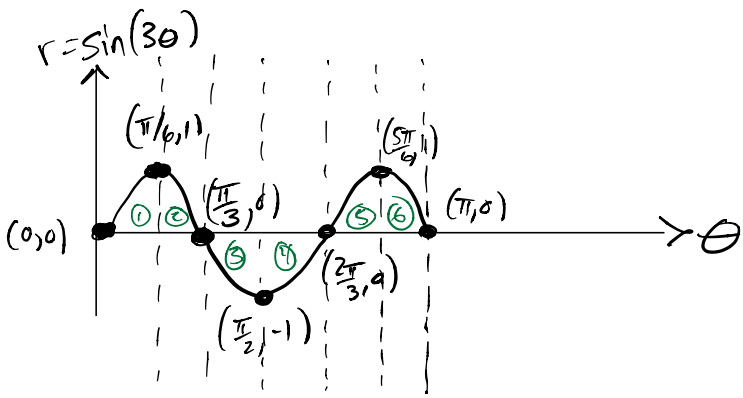
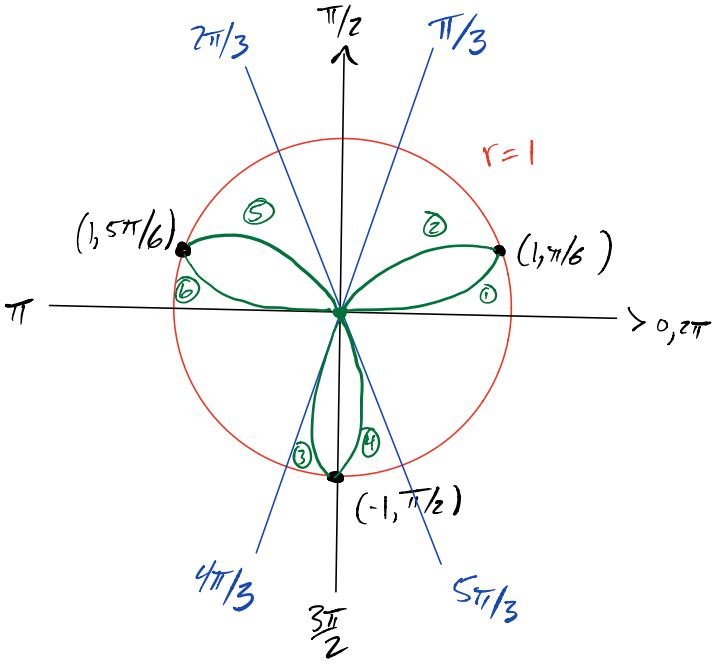
$$x = t^2,\ y = \ln(t)$$

then sketch the curve and indicate with an arrow the direction in which the curve is traced as t increases.

$\ln(x) = \ln(t^2) = 2\ln(t) = 2y$

$\Rightarrow y = \frac{1}{2}\ln(x), \quad 0 < t$



Polar Coordinates

6. Sketch the rose $r = \sin(3\theta)$, $0 \le \theta \le \pi$. Label the tips of the petals and draw arrows to indicate the direction in which the curve is traced.